

of the Web page, the extensive API provides several methods and events to let scripts control the playback of the media. The simplest methods to use are the *play()*, *pause()*, and setting *currentTime* to rewind to the beginning.

Drag and drop

Drag and drop is one of the most fundamental interactions afforded by graphical user interfaces. In one gesture, it allows users to pair the selection of an object with the execution of an action, often including a second object in the operation. It's a simple yet powerful UI concept used to support copying, list reordering, deletion (*a la* the Trash/Wastebin), and even the creation of link relationships.

Since it's so fundamental, offering drag-and-drop in Web applications has been a no-brainer ever since browsers first offered mouse events in DHTML (Dynamic HTML). But, although *mousedown*, *mousemove*, and *mouseup* made it possible, the implementation has been limited to the bounds of the browser window. Additionally, since these events refer only to the object being dragged, there's a challenge to find the subject of the drop when the interaction is completed.

Of course, that doesn't prevent most modern JavaScript frameworks from abstracting away most of the problems and throwing in some flourishes while they're at it. But, wouldn't it be nice if browsers offered first-class support for drag-and-drop, and maybe even extended it beyond the window sandbox?

As it turns out, this very wish is answered by the HTML 5 specification section on new drag-and-drop events, and Firefox 3.5 includes an implementation of those events.

The latest events

The latest drag and drop events specified for HTML 5 are:

- **dragstart** – A drag has been initiated, with the dragged element as the event target.

- **drag** – The mouse has moved, with the dragged element as the event target.
- **dragenter** – The dragged element has been moved into a drop listener, with the drop listener element as the event target.
- **dragover** – The dragged element has been moved over a drop listener, with the drop listener element as the event target. Since the default behaviour is to cancel drops, returning *false* or calling *preventDefault()* in the event handler indicates that a drop is allowed here.
- **dragleave** – The dragged element has been moved out of a drop listener, with the drop listener element as the event target.
- **drop** – The dragged element has been successfully dropped on a drop listener, with the drop listener element as the event target.
- **dragend** – A drag has been ended, successfully or not, with the dragged element as the event target.

For a detailed list and for explanations, read www.whatwg.org/specs/web-apps/current-work/#dnd.

Consider the following example of jQuery:

```
<div id="newschool">
  <div class="dragne">Drag me!</div>
  <div class="drophere">Drop here!</div>
</div>

<script type="text/javascript">
$(document).ready(function() {
  $('#newschool .dragne')
    .attr('draggable', 'true')
    .bind('dragstart', function(ev) {
      var dt = ev.originalEvent.

dataTransfer:

      dt.setData('Text', 'Dropped in zone!');
      return true;
    })
    .bind('dragend', function(ev) {
      return false;
    });
  $('#newschool .drophere')
    .bind('dragenter', function(ev) {
      $(ev.target).addClass('dragover');
    })
    .bind('dragleave', function(ev) {
      return false;
    })
    .bind('dragover', function(ev) {
      $(ev.target).removeClass('dragover');
      return false;
    })
    .bind('drop', function(ev) {
      var dt = ev.originalEvent.

dataTransfer:

      alert('dt: ' + dt.getData('Text'));
      return false;
    });
});
</script>
```

```
return false;
})
.bind('dragleave', function(ev) {
  $(ev.target).removeClass('dragover');
  return false;
})
.bind('dragover', function(ev) {
  return false;
})
.bind('drop', function(ev) {
  var dt = ev.originalEvent.

dataTransfer:

  alert('dt: ' + dt.getData('Text'));
  return false;
});
});
</script>
```

Since a detailed explanation of the features and demonstrations is out of the scope of this article, you can always refer to a lot of interesting content on:

- https://developer.mozilla.org/En/DragDrop/Drag_and_Drop
- <http://decafbad.com/blog/2009/07/15/html5-drag-and-drop>
- <http://ljoanneau.com/lab/html5/demodragdrop.html>

Last, but definitely not the least

If you fancy getting more involved, there's still time—try using HTML 5 and give your feedback to the specification group via the WHATWG mailing lists [www.whatwg.org/mailling-list]. The editor, Ian Hickson, has put out a call for people to review the specs—looking for confusing items, typos, and other small problems. If you find one, you'll be mentioned in the acknowledgements. There's plenty to do, so dive in!  **END**

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An avid Twitter user and a social media enthusiast, the author is a passionate blogger and a professional gamer too. He also feels compelled to be opinionated about anything that comes his way, be it Linux distributions, our marketing strategies, table etiquettes or even the fabled Ramsay movies!